8th sem (Regular & Back)
SC IT-815
(CSE, IT)

SPRING END SEMESTER EXAMINATION-2014

8th Semester B.Tech

SOFT COMPUTING IT-815

(Regular-2010 & Back of Previous Batches)

Full Marks: 60 **Time: 3 Hours**

Answer any SIX questions including Question No.1 which is compulsory.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. a) Differentiate between soft computing and hard computing and also between AI & CI. [2 × 10

   b) Briefly explain five soft computing characteristics.

   c) Draw the schematic diagrams of an expert system and an intelligent system.

   d) Explain three types of universes of discourse with examples.

   e) Explain composite and non-composite MFs with suitable example.

   f) Explain any two types of complement operators with analytical expression and diagrams.

   g) Define four of the most frequently used T-norm operators and S – norm operators.

   h) Explain binary fuzzy relation and cylindrical extension of one-dimensional fuzzy sets.

i) Explain perceptron learning rules.

j) Explain elitism in GA through example.

2. a) Define the following terms associated with fuzzy sets: [4
(i) Support, (ii) Core, (iii) Normality, (iv) Convexity,
(v) Crossover points, (vi) α -cut & strong α -cut, (vii) Bandwidth,
(viii) Open left, open right, closed, (ix) Symmetry.

b) Discuss with necessary analytical expressions and diagrams the following 1-D Parameterized MFs: [4
(i) Triangular MF, (ii) Trapezoidal MF, (iii) Gaussian MF, and
(iv) Bell MF

3. a) Define linguistic variable as a quintipule and also define the following terms: [4
(i) CON (A), (ii) DIL (A), (iii) INT (A) and (iv) DIM (A)

b) Construct the MFs for following composite linguistic terms with the meaning of primary terms 'young' and 'old' defined by the following MFs: [4

$\mu_{young}(x) = bell(x; 20, 2, 0)$

$\mu_{old}(x) = bell(x; 30, 3, 100)$

(i) Old but not too old

(ii) Not very young and not very old

a) Explain diagrammatically the fuzzy reasoning for following situations. (Assume Mamdani's fuzzy implication and max-min composition): [4
(i) Single rule with antecedent
(ii) Single rule with multiple antecedents
(iii) Multiple rule with multiple antecedents.

b) Explain with neat diagrams the Mamdani fuzzy model using *min* as T-norm operator and *max* as T-conorm operator. Explain five defuzzification strategies. [4

5. a) Explain four steps of the process of fuzzy reasoning (or approximate reasoning) with a suitable block schematic diagram of FIS in general. [4

b) What are the four basic requirements for T-norm operators as well as for S-norm operators? Define four T-norm operators and four S-norm operators. [4

6. Enumerate the characteristics of derivative free optimization. Explain the concept of GA and its major components. Write down the basic steps of GA. [8

7. a) What is the Compositional Rule of Inference? Derive the expression B=A o F, where A, B and F have their usual meanings. [4

b) Explain Perceptron with diagram and explain how its limitation is handled by using MLP. [4

8. Writes short notes on any two: [4 × 2

a) Principle of RBFN.

b) Hebbian Learning.

c) ADALINE model.

d) ANFIS architecture.

xxxxx